10.8 Sample Programs to illustrate classes and objects

Program 1: Write a program to calculate area and circumference of a circle

```
class Circle:
        pi=3.14
def __init__(self,radius):
        self.radius=radius
def area(self):
        return Circle.pi*(self.radius**2)
def circumference(self):
        return 2*Circle.pi*self.radius
r=int(input("Enter Radius: "))
C=Circle(r)
print("The Area =",C.area())
print("The Circumference =", C.circumference())
```

Output:

```
Enter Radius: 5
The Area = 78.5
The Circumference = 31.400000000000002
```

Program 2: write a menu driven program that keeps record of books available in you school library

```
class Library:
        def __init__(self):
                self.bookname=""
                self.author=""

        def getdata(self):
                self.bookname = input("Enter Name of the Book: ")
                self.author = input("Enter Author of the Book: ")

        def display(self):
                print("Name of the Book: ",self.bookname)
                print("Author of the Book: ",self.author)
                print("\n")
```


XII Std - CS EM Chapter-10.indd 177 24-01-2020 12:42:36